Protocol for the Evaluation of Local Anaesthesia in the endovascular repair of ruptured abdominal aortic aneurysms: a Target trial and Economic evaluation (ELATE) study

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Abstract: 454 words

Main text: 3728

Tables: 2

Figures: 1

References: 43

Summary of protocol amendments

Version 1.0: Original protocol

Version 2.0: Incorporates Amendment 1 (dated 11/12/2025)

Version 3.0: Incorporates Amendment 2 (dated 08/04/2026)

Amendment 1 (11/12/2025)

Section reference	Original wording	Revised wording (changes bolded)	Rationale
Ethics and Dissemination	-	Patients who have opted-out of their data being used for research via the National Data Opt-Out will be excluded from the ELATE study. More information on the National Data Opt-Out is available here: https://digital.nhs.uk/services/national-data-opt-out	Addition of opt-out information requested by CAG

Amendment 2 (08/04/2026)

Section reference	Original wording	Revised wording (changes bolded)	Rationale
Author list	-	-	Author list added.
Abstract	How effective and cost-effective is local infiltrative anaesthesia versus general anaesthesia for endovascular repair of ruptured abdominal aortic aneurisms (rAAA)?	**What is the clinical and cost-effectiveness of** local infiltrative anaesthesia versus general anaesthesia for endovascular repair of ruptured abdominal aortic aneurysms (rAAA)?	Research question re-worded for easier communication.
Eligibility criteria	People captured in the NVR for non-elective endovascular repair for rAAA in hospitals in England who receive either local or general anaesthesia will be eligible for inclusion in the TTE. Exclusions will include thoracoabdominal/thoracic aneurysm, being aged <50 years, and being recorded as receiving complex or	People captured in the NVR **between 1 January 2014 to 31 December 2024** for non-elective endovascular repair for rAAA in hospitals in England who receive either local or general anaesthesia will be eligible for inclusion in the TTE. ... **We will also exclude people who are hospitalised at hospitals where there is a low volume of cases (those meeting the eligibility criteria), as required by the**	Timeframe for study inclusion and low volume hospital exclusion added as agreed by investigators.

	revision endovascular repair.	**instrumental variable analysis (see Statistical analysis). The 'low volume' definition (e.g., 10 cases per year) will be determined after conducting descriptive analyses and before any comparative effectiveness analyses.**	
Treatment strategies	-	**... We will consider scenarios where the procedure changes from using local to general anaesthetic in a sensitivity analysis (see Causal contrasts).**	Investigators agreed that a sensitivity analysis is likely necessary to account for those who switch from local to general anaesthetic during surgery.
Causal contrasts	We will compare those who are reported as receiving either local or general anaesthetic as part of EVAR for rAAA. It is possible that people may start with local anaesthetic, but are switched to general anaesthetic during the procedure. This would likely be recorded as general anaesthetic in the NVR database. Thus, we will estimate the per-protocol effect rather than the observational analogue of the intention-to-treat (ITT) effect.	We will compare those who are reported as receiving either local or general anaesthesia as part of EVAR for rAAA. **We anticipate approximately 10-20% of people to convert from local to general anaesthetic due to unmeasured factors such as pain tolerance and patient stress/fear.** This would likely be recorded as general anaesthetic in the NVR database**, or recorded as both local and general anaesthetic from Jan 2019 on since this data item was changed from single response to multiple response at that time.** Thus, we will estimate the per-protocol effect rather than the observational analogue of the intention-to-treat (ITT) effect. **We will consider a sensitivity analysis restricting to the population hospitalised post-2019, for whom the initial intention to use local anaesthetic (if both local and general anaesthetic are selected) is available.**	Investigators agreed that a sensitivity analysis is likely necessary to account for those who switch from local to general anaesthetic during surgery.
Statistical analysis	A range of data-driven approaches have been proposed to identify subgroups.[34-39] We will use the 'Virtual Twins' approach,[34] which constructs a decision tree	**We will consider different data-driven approaches to identify subgroups for which there is treatment effect heterogeneity,[34-40] although the study sample size may**	It was agreed among investigators that data-driven methods to identify subgroups are likely

	that groups individuals based on characteristics predictive of expected treatment benefit from local versus general anaesthesia. We will also use qualitative interaction trees[35] to identify the specific groups of patients for whom local anaesthesia is expected to be better, worse, or similar than general anaesthesia. To guard against overfitting/false discovery, we will use an 'honest' estimation approach.[40]	**limit our ability to conduct these analyses.**	to be under-powered.
Patient and Public Involvement	Building on the ESORT-V study, …	Building on the ESORT-V study **(NIHR202788),[41]** …	Reference added to precedent research project.
Ethics and Dissemination	We will apply for research ethics permissions to the NHS Research Ethics Committee and the London School of Hygiene & Tropical Medicine Ethics Committee. We will also require Confidentiality Advisory Group Section 251 approval for the use of patient data without individual consent for research.	**This study was approved by the NHS East Midlands – Leicester South Research Ethics Committee (REC reference 25/EM/0263) and the London School of Hygiene & Tropical Medicine Ethics Committee (reference 33138). The Confidentiality Advisory Group (CAG) has given advice to the Health Research Authority under section 251 of the NHS Act 2006 to support this research (CAG reference 25/CAG/0153).**	Ethics approvals received between publication of protocol v2 and v3 updated.

Abstract

Research question: What is the clinical and cost-effectiveness of local infiltrative anaesthesia versus general anaesthesia for endovascular repair of ruptured abdominal aortic aneurysms (rAAA)?

Background: For people who have surgery for rAAA, outcomes are poor; even after endovascular surgery, 40% die within one year. National Institute for Health and Care Excellence (NICE) recommend that surgeons 'consider' local anaesthesia for endovascular repair of rAAA. However, no rigorous evidence exists to inform this decision. There is wide variation in the type of anaesthesia used across United Kingdom (UK) vascular centres. A randomised controlled trial (RCT) to address this question is likely impractical and unethical; thus, high quality observational evidence is needed.

Aims and objectives: To evaluate the effectiveness and cost-effectiveness of local versus general anaesthesia for endovascular repair (EVAR) of rAAA. The objectives are:

1. To assess the effectiveness and cost-effectiveness of local versus general anaesthesia for EVAR of rAAA overall.

2. To assess the effectiveness and cost-effectiveness of local versus general anaesthesia in patient subgroups (e.g. sex, frailty).

3. To provide recommendations on the choice of anaesthesia according to patient subgroup.

Methods: We will provide rigorous evidence using the target trial emulation framework on the clinical and cost-effectiveness of local versus general anaesthesia for EVAR of rAAA. Our target trial emulation will use observational data from the National Vascular Registry (NVR) linked to Hospital Episode Statistics (HES) and Office of National Statistics (ONS) mortality data. It will apply criteria to define the study populations, intervention and comparator strategies and outcomes of direct relevance to clinical decision-making. Criteria will be informed by panels of clinicians and patients/public representatives. Based on advice from patients and clinicians, the primary outcome will be 30-day mortality. Secondary outcomes will include days alive and out of hospital.

We will use an instrumental variable analysis, which can provide accurate estimates of treatment effectiveness, even when there are unmeasured differences between the intervention and comparator groups. The cost-effectiveness analyses will be based on resource use and mortality data from the NVR-HES-ONS data. We will report relative clinical and cost-effectiveness according to pre-specified patient subgroups.

To help ensure this study generates evidence that directly meets the needs of decision-makers, we will convene panels of clinicians, patient/public representatives, and other

stakeholders at the study outset. These groups will help translate results to directly inform clinical decision-making including updates to clinical guidelines for rAAA.

Anticipated impact and translation: Our clinical and patient/public panels and 'all interest holder event' will help ensure the research is directly relevant to the National Health Service (NHS). Our clinical co-investigators and steering committee will help translate findings to national decision-makers. This study will exemplify how timely input from key stakeholders can help ensure a target trial emulations directly inform clinical practice.

Introduction

Ruptured abdominal aortic aneurysm (rAAA) is almost always fatal without emergency intervention by open surgery or minimally invasive endovascular repair (EVAR). The National Institute for Health and Care Excellence (NICE) recommend EVAR rather than open repair for rAAA, noting that EVAR provides more benefit than open surgical repair for most people, especially men over 70 and women of any age.[1] This recommendation is based on results from the IMPROVE (Immediate Management of the Patient with Rupture: Open Versus Endovascular repair) trial that compared EVAR with open repair.[2] However, outcomes from emergency EVAR remain poor; around 20% of people die in hospital[3] and 40% die within one year.[4]

NICE also recommends that surgeons 'consider using local anaesthesia alone for people having EVAR of a ruptured AAA'.[1] This recommendation is based on a post-hoc non-randomised comparison of local versus general anaesthesia from the EVAR arm of the IMPROVE trial,[2] which included a small sample of 142 patients and is subject to confounding by indication.

There is considerable variation across UK centres in the use of local versus general anaesthesia in rAAA, with a median of 40% using local anaesthesia but with an inter-quartile range of 23-60%.[5] This variation highlights that for patients having EVAR of rAAA, use of either local or general anaesthesia is likely to depend on the preference of the local clinical team as well as the clinical characteristics of the patient. This wide variation in practice is unlikely to change in the persistent absence of rigorous evidence on the relative clinical and cost-effectiveness of alternative anaesthetic approaches both overall, and for specific patient subgroups.

There is a lack of RCT evidence on the comparative effectiveness of local versus general anaesthesia for endovascular repair of rAAA. A 2023 Cochrane review found no published or ongoing RCTs comparing general versus loco-regional anaesthesia in elective or emergency endovascular repair for abdominal aortic aneurysm.[6] A 2019 systematic review found 4 non-randomised studies comparing local with general anaesthesia in emergency endovascular repair for rAAA.[7] All four studies reported lower 30-day mortality following local versus general anaesthesia (risk ratios between 0.40 and 0.68). Authors suggest that poorer outcomes with general anaesthesia may be due to relaxation of tissues and release of haemostatic tamponade, and the significant haemodynamic effects of general anaesthesia, including loss of vascular tone.[5] These may be exacerbated in patients with rAAA experiencing shock, a sudden episode of instability and low blood pressure precipitating myocardial infarction, stroke or end organ damage. General anaesthesia may also cause delays in conducting EVAR. However, confounding was a major concern in these observational studies.

A more recent meta-analysis by Deng et al that includes 8 observational studies reported similar results that local anaesthesia is associated with lower mortality than general anaesthesia.[8] However, this study raised the concern that local anaesthesia has some potential disadvantages, including patient anxiety, discomfort, and movement/restlessness caused by pain or spontaneous breathing, which may increase the urgency and difficulty of the endovascular repair. General anaesthesia is the more familiar mode of anaesthesia and may also help attenuate ischaemia reperfusion injury.

The possibility that relative effectiveness of local versus general anaesthesia differs by subgroup is raised by the IMPROVE trial of endovascular versus open repair for rAAA[9] and the Deng et al review, which suggests there may be differential effects according to age and systolic blood pressure. More generally, there is systematic review evidence in elective endovascular versus open repair showing worse outcomes for females.[10]

Previous studies evaluating local vs general anaesthesia with EVAR have not formally evaluated health-related quality of life (QoL), costs or cost-effectiveness. Of the 5 studies identified evaluating EVAR versus alternative strategies (open surgery or medical management), only the IMPROVE trial considered QoL, costs and cost-effectiveness.[11]

Published guidelines reflect the potential benefits of local anaesthesia, but also concerns about the quality of the evidence. European Society for Vascular Surgery guidelines recommend local anaesthesia for endovascular repair of rAAA whenever tolerated by the patient.[12] NICE guidelines include the recommendation to “consider using local infiltrative anaesthesia alone for people having EVAR of a rAAA” to raise awareness of the option.[1]

Qualitative research shows that decision-making in the mode of anaesthesia in emergency surgery (including EVAR of rAAA) is complex. Inter-linking themes include professional expertise, patient and professional preferences, practicalities, norms, and policies.[13] Discrepancies in mode of anaesthesia were mirrored by differences in the information presented to patients and the level of involvement patients had in their care. The study highlighted the lack of evidence to inform individual-level decision-making. In response, this study will generate real-world evidence to support clinicians in tailoring their anaesthetic approach to individual patient needs.

Aims and Objectives

We aim to evaluate the effectiveness and cost-effectiveness of local versus general anaesthesia for endovascular repair of rAAA. The specific study objectives are:

1. To assess the overall effectiveness and cost-effectiveness of local versus general anaesthesia for EVAR of rAAA.

2. To assess the effectiveness and cost-effectiveness of local versus general anaesthesia in patient subgroups (e.g. according to age, sex, frailty, systolic blood pressure, pre-existing long-term conditions).
3. To provide recommendations on the choice of anaesthesia according to the patient subgroup.

Methods and analysis

Study design and data sources

We will use the target trial emulation (TTE) framework[14,15] to evaluate the effectiveness and cost-effectiveness of local versus general anaesthesia for endovascular repair of rAAA in a cohort study using observational data.

The study will use linked data from the National Vascular Registry (NVR),[16] Hospital Episode Statistics (HES),[17] and Office for National Statistics (ONS) mortality[18] (NVR-HES-ONS linked data) on patients undergoing endovascular repair for rAAA. We will collate information on diagnosis, case-mix, surgical procedures received, resource use and outcomes using these data.

Target Trial Emulation

We will emulate a hypothetical pragmatic RCT for assessing the clinical and cost-effectiveness of local versus general anaesthesia by specifying the target trial's protocol and its corresponding emulation, following the TARGET reporting guidelines.[19] The TTE is summarised in **Table 1**.

Eligibility criteria

People captured in the NVR between 1 January 2014 to 31 December 2024 for non-elective endovascular repair for rAAA in hospitals in England who receive either local or general anaesthesia will be eligible for inclusion in the TTE. Exclusions will include thoracoabdominal/thoracic aneurysm, being aged <50 years, and being recorded as receiving complex or revision endovascular repair. We will also exclude people who are hospitalised at hospitals where there is a low volume of cases (those meeting the eligibility criteria), as required by the instrumental variable analysis (see Statistical analysis). The 'low volume' definition (e.g., 10 cases per year) will be determined after conducting descriptive analyses and before any comparative effectiveness analyses.

Treatments strategies

We will compare local anaesthesia, defined by record of local anaesthetic or infiltration, with general anaesthesia, recorded as general anaesthetic, using the anaesthetic type data item in the NVR database. We will consider scenarios where the procedure changes from using local to general anaesthetic in a sensitivity analysis (see Causal contrasts).

Assignment procedures and follow-up

Time zero refers to the timepoint of randomisation/treatment assignment in a TTE. At time zero, all the participant eligibility criteria are met, treatment assignment happens, and the follow-up time begins. We cannot randomise as we would in the target trial. Here, we will define time zero as the date of the endovascular repair recorded in the NVR, at which the anaesthesia (treatment) would be delivered. We will triangulate this date with the corresponding data from HES on finished consultant episode and Operating Procedure Codes Supplement (OPCS) procedure dates. Follow-up will begin on this date.

Outcomes

The primary outcome is 30-day all-cause mortality defined using the death date recorded in the ONS data. We will draw on our patient and public involvement (PPI) panel to determine the relative importance of the secondary clinical outcomes, which will include in-hospital mortality, 1-year mortality, days alive and out of hospital at 30 days and 1-year, discharge destination, complications, and readmissions or reinterventions within 30 days and 1-year. We will consider adding any secondary outcomes the PPI panel suggest which are not already included and update the protocol accordingly. DAOH can be calculated in 2 different ways: the first includes the number of days out of hospital for those who die before the end of follow-up; the second assigns zero DAOH for these who die during follow-up. We will use the first DAOH measure in the main analysis and the second in a sensitivity analysis.

We will investigate several health economic outcomes as part of the cost-effectiveness analysis, including total costs, length of stay, incremental quality adjusted life years (QALYs), and incremental net monetary benefit.

Causal contrasts

We will compare those who are reported as receiving either local or general anaesthesia as part of EVAR for rAAA. We anticipate approximately 10-20% of people to convert from local to general anaesthetic due to unmeasured factors such as pain tolerance and patient stress/fear. This would likely be recorded as general anaesthetic in the NVR database, or recorded as both local and general anaesthetic from Jan 2019 on since this data item was changed from single response to multiple response at that time. Thus, we will estimate the per-protocol effect rather than the observational analogue of the intention-to-treat (ITT) effect. We will consider a sensitivity analysis restricting to the population hospitalised post-2019, for whom the initial intention to use local anaesthetic (if both local and general anaesthetic are selected) is available.

Identifying assumptions

The ‘target trial’ and the TTE make the assumptions of exchangeability (the treatment groups are comparable on average) and positivity (patients have a non-zero probability of

being allocated to local or general anaesthetic). The 'target trial' meets these assumptions via randomisation to local versus general anaesthetic. However, the TTE cannot randomise since we are using observational data. Thus, we will use an instrumental variable analysis,[20] which avoids making the common assumption of conditional exchangeability (on measured confounders). Rather, we will use the exogenous variation in the use of local versus general anaesthetic across vascular centres in England as an instrument to conduct the instrumental variable analysis.

The instrumental variable is a vascular centre's tendency to prefer local to general anaesthesia for endovascular repair following rAAA. This definition of the instrumental variable at the level of the 'centre' rather than the individual vascular surgeon, recognises the importance of the whole surgical and perioperative team in recommending the mode of anaesthesia, and the potential role of 'centre-effects' in driving preferences for either mode of anaesthesia.

We will define the tendency to prefer local to general anaesthesia from NVR data by calculating the proportions of rAAA cases who have local versus general anaesthesia for rAAA at the same centre for each individual.[4] We will undertake initial analysis to examine within-centre variation over time in the percentage of patients having EVAR under local anaesthesia to inform the appropriate 'look back' period for the instrumental variable (6-, 12-, or 24- months).

To be valid, the instrument must meet 4 main assumptions, which can only be partially tested. These assumptions are described in **Table 2**. We will conduct the appropriate analyses to partially test these assumptions as part of our statistical analysis. This study follows others which have successfully used variation in vascular surgery provision as an instrument to infer causal treatment effects.[21-23]

Statistical analysis

Clinical effectiveness

We will first describe the study population to understand how patient and context-level characteristics differ by treatment group. We will collate information from linked NVR-HES-ONS data on potential confounders of the causal association between anaesthetic type and the study outcomes (**Figure 1**). These will include patient characteristics (age, sex, ethnicity, index of multiple deprivation), relevant prognostic indicators, markers of rAAA severity, comorbidities,[24] body-mass index, and frailty.[25,26]

While our eligibility criteria will restrict the study population to that for which there is clinical uncertainty as to whether local versus general anaesthetic is better, we expect differences in measured and unmeasured confounders between the treatment groups. Thus, we will use an instrumental variable analysis to address risks of residual confounding,[20] for example because the patients who receive local anaesthesia might

have a better prognosis to those who receive general anaesthesia according to variables that are not recorded in routine data, such as a patient's haemodynamic stability.

An instrumental variable must be associated with receipt of the intervention, but not have a direct effect on the outcome (**Table 2**). In this study, our preference-based instrumental variable, a vascular centre's tendency to use local anaesthesia, will build on previous work on preference-based 'tendency to operate' instrumental variables in emergency surgical patients.[21-23] This measure differs across vascular centres,[5] but is unlikely to be correlated with quality of care, after conditioning on past outcomes, and therefore meets the main criteria required for a valid instrumental variable.

We will estimate the effect of local versus general anaesthesia on each outcome. Probit regression models will be used to estimate the initial propensity score (first stage), conditional on the measured potential confounders. Generalised Linear Models with appropriate link functions will be used for continuous or binary outcomes (second stage). Models at both stages will adjust for case-mix, organisational- and temporal factors described above including time-period, proxies for care quality, defined by rates of 30-day and 1-year mortality for each NHS hospital trust following endovascular repair for rAAA for the year prior to the admission in question. The estimates will be reported with bootstrapped confidence intervals that allow for the clustering of individuals within NHS trusts and NHS region.

We will estimate the relative effectiveness and cost-effectiveness of local versus general anaesthesia according to patient subgroups. The clinical panels will refine the list of a priori patient characteristics that, according to the literature, may modify the relative effectiveness of local versus general anaesthesia in endovascular repair of rAAA. We will consult with our clinical and PPI panels to refine the subgroups of interest. In this protocol, we pre-specify the following subgroups: sex, age group, frailty (measured using the hospital frailty risk score[26]), systolic blood pressure at baseline, and the prevalence of other long-term conditions (e.g. chronic lung disease, chronic kidney disease, ischemic heart disease), past medical history (e.g. current smoker or not) and case severity according to the modified form of the Hardman index which is available from the NVR. The Hardman index for predicting the risk of an adverse outcome is not captured in the NVR. However, four of the five factors used to derive the index are included. Therefore, a modified Hardman index will be derived based on age, haemoglobin, serum creatinine, and ECG findings only (excluding data on loss of consciousness).

Cost-effectiveness

We will assess the relative costs and cost-effectiveness of local versus general anaesthesia for endovascular repair following rAAA. This cost-effectiveness analysis (CEA) will take a hospital and personal health services perspective[27] to report results over a one-year time horizon. The IMPROVE trial suggested that by one-year after rAAA, Quality

of Life (QoL), and mortality is similar to the age- and gender matched general population. The CEA will combine patient-level resource use (NVR and HES) and mortality data (ONS), with estimates from the literature for QoL, community care, and unit costs.[28] The CEA will consider whether potential gains from local anaesthesia in reduced mortality and shorter initial hospital stay are offset by any increased risks of reinterventions and readmissions, especially amongst patients at higher levels of baseline risk.

Measurement of resource use, unit costs and outcomes:

The study will extract resource use data from linked NVR-HES data for the index admission and all readmissions. Unit costs will be taken from a literature review, the NHS payment by results database, and the PSSRU unit cost database.[28] For the base case analysis we will combine resource use measures with unit costs to report total costs per patient up to one year.

We will extract information on QoL from a review of the literature, including the QoL estimates from the IMPROVE trial.[29] In the base case, we will report quality-adjusted life years (QALYs) up to one-year by combining survival time with QoL estimates using the 'area under the curve' approach, adjusting for age and sex.

Analysis

We will extend the previous instrumental variable analysis to report the incremental QALYs and costs of local versus general anaesthesia. We will report the relative cost-effectiveness according to incremental net monetary benefit. We will test whether conclusions are robust to key assumptions, in particular, the validity of the instrumental variable, to alternative perspectives (hospital versus personal social services), to alternative sources for the QoL and unit cost data, and to different time horizons (lifetime vs 1-year). We will access NVR-HES-ONS linked mortality and resource use data after the initial admission to inform the approach to survival extrapolation in reporting lifetime cost-effectiveness.

Missing and miscoded data

Missing or unreported ethnicity data in eligible admissions (~10%), and deprivation (~1%) will be minimised by using ethnicity and deprivation data from patients' other linked episodes in the NVR-HES-ONS linked data. HES data for age and sex are almost complete. We will recognise the potential impact of improvements in coding over time by including a covariate for financial year in the analysis models. For any missing covariate data, we will make plain any assumptions that we do make. For example, for baseline covariates with missing values, we will examine whether this pattern of missingness differed across the intervention and comparator groups. We will use the approach recommended by Rosenbaum to making minimal, but transparent assumptions about

the missing data while balancing patterns of missingness across intervention and control groups.[30]

Sensitivity analyses

We will undertake extensive sensitivity analyses including: exclusion of cases where there is doubt about clinical coding, alternative specifications of the instrumental variable, alternative assumptions about the approach to missing data (including approaches that invoke the 'missing at random assumption' rather than the 'complete case assumption'), and methods that make different assumptions about confounding including double-robust methods.

We will draw on the findings from objectives 1 and 2 to recommend which patient groups should have local versus general anaesthesia, according to relative effectiveness and cost-effectiveness. To do so we will estimate individual level effects on outcomes (e.g., clinical effectiveness, costs, Incremental Net Benefit) by contrasting actual and predicted counterfactual outcomes obtained using the local instrumental variable approach.[31,32] These effects will then be aggregated for a limited set of pre-specified subgroups informed by Clinical and PPIE advisers.[33] We will consider different data-driven approaches to identify subgroups for which there is treatment effect heterogeneity,[34-40] although the study sample size may limit our ability to conduct these analyses. Confidence intervals for all subgroup effects will be obtained by bootstrapping.

Sample size calculation

A difference in the absolute risk of the primary outcome (30-day mortality) of 10%, is judged important to patients and for informing decision-making.[4]

We assume a population standard deviation for mortality of 0.3, that 50% of patients have local anaesthesia, and with assumed compliance rate between the IV and treatment of 50%, 60% and 70%. This implies required sample sizes of 1130, 785 and 577 respectively, at 80% power.[20] Our previous work with the NVR and published NVR data on EVAR for rAAA over the proposed timeframes suggests that the anticipated sample size is likely to be ~1,500. As there will be minimal missing data for 'all cause' death, the sample size will be sufficient even if there is only moderate correlation of the instrumental variable with anaesthesia type.

Patient and Public Involvement

Patients and the public have reiterated the importance of this research. In 2 panels held in 2023, contributors emphasised that stronger evidence comparing local and general anaesthesia for rAAA could reduce patient mortality and address regional inequalities in care. They endorsed the TTE approach, noting they would not wish to participate in an

RCT comparing anaesthesia types. They supported the use of routine NHS data to improve clinical decision-making and consistency across centres.

The study design follows best practice for achieving impact through early and inclusive stakeholder engagement. Building on the ESORT-V study (NIHR202788),[41] our team has brought together vascular surgeons, anaesthetists, specialist nurses, patients, NICE guideline developers, and NHS commissioners. Their input will help generate practical recommendations for anaesthesia provision, improving patient outcomes, shortening hospital stays, and reducing healthcare costs. The approach aligns with the Goldacre review guidance[42] and NICE standards for the generation of real-world evidence.[43] These approaches will exemplify best use of NHS routine data to tailor interventions to patient needs.

PPI will inform the study throughout. Two design workshops, each with approximately 10 participants, will review patient-facing materials and contribute to protocol design, interpretation, and dissemination. Participants, recruited from national vascular and NIHR advisory panels, will reflect diverse backgrounds and lived experience. Workshops will explore outcomes important to patients, including mortality, discharge destination, and days alive and out of hospital, as well as complications and readmissions, ensuring the study reflects patient priorities.

Ethics and Dissemination

Ethics

This study was approved by the NHS East Midlands – Leicester South Research Ethics Committee (REC reference 25/EM/0263) and the London School of Hygiene & Tropical Medicine Ethics Committee (reference 33138). The Confidentiality Advisory Group (CAG) has given advice to the Health Research Authority under section 251 of the NHS Act 2006 to support this research (CAG reference 25/CAG/0153). Patients who have opted-out of their data being used for research via the National Data Opt-Out will be excluded from the ELATE study. More information on the National Data Opt-Out is available here: https://digital.nhs.uk/services/national-data-opt-out

Dissemination

We will work with our multidisciplinary study team to disseminate results through various channels to maximise the impact of this study.

We will publish study findings in peer-reviewed scientific journals and present the results at national and international conferences. We will have an all-stakeholder online event to help maximise the reach of the findings and to ensure they feed directly and quickly into clinical practice. The workshop will be co-led by our PPIE co-investigators, clinical co-investigators, and the study co-PIs, and include clinicians (vascular surgeons,

anaesthetists), patients, professional bodies, NICE guidelines developers, and NHS commissioners. We will discuss the direct implications of the findings for practice, and any barriers to changing practice at the local level. We will discuss the implications of findings for NICE guidelines on rAAA and other emergency endovascular aortic procedures, e.g. ruptured thoracic aneurysm or aortic dissection.

We will also hold a PPI translation workshop where we will consider how the study findings can improve the information available to patients and relatives. We will discuss the priorities for future research provoked by this study's findings.

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Table 1. Target Trial Emulation (TTE) specification for the Evaluation of Local Anaesthesia in the endovascular repair of ruptured abdominal aortic aneurysms: a Target trial and Economic evaluation (ELATE) study

Protocol component	Target trial specification	Target trial emulation
Eligibility criteria	**Include:** - People hospitalised for non-elective endovascular repair for rAAA in England **Exclude***:* - Thoracoabdominal/thoracic aneurysm - Aged <50 years - Complex or revision endovascular repair	**Include:** - People hospitalised for non-elective endovascular repair for rAAA in England 1 January 2014 to 31 December 2024 captured in the NVR **Exclude:** - Thoracoabdominal/thoracic aneurysm - Aged <50 years - Complex or revision endovascular repair - People who receive other types of anaesthesia neuraxial block (spinal/epidural) - Those hospitalised at hospitals with a low volume of cases meeting the inclusion/exclusion criteria per year
Treatment strategies	**Intervention** - Local anaesthesia for endovascular repair for rAAA **Comparator** - General anaesthesia for endovascular repair for rAAA	**Intervention** - Local anaesthesia for endovascular repair for rAAA measured as local infiltration record in NVR **Comparator** - General anaesthesia for endovascular repair for rAAA measured in the anaesthetic type variable in the NVR
Assignment procedures	Individuals are randomly assigned to anaesthesia type at hospital admission (start of operation).	Randomisation is not possible using observational data. Anaesthesia type is observed using the NVR data.
Follow-up	Follow-up start (time 0) is synchronous with study entry and randomisation (the surgery date). Follow-up ends at the earliest of the outcome ascertainment window (see outcomes), death, or end of study period.	Follow-up start (time 0) is synchronous with study entry and treatment assignment (the surgery date). Follow-up ends at the earliest of the outcome ascertainment window (see outcomes), death, or end of study period.
Outcomes	*Primary outcome:* - 30-day all-cause mortality *Secondary outcomes:*	*Primary outcome:* - 30-day all-cause mortality *Secondary outcomes:*

	- In-hospital all-cause mortality - 1-year all-cause mortality - Days alive and out of hospital at 30-days and 1-year - Discharge destination - Complications - Readmissions or reinterventions within 30-days and 1-year - Incremental QALY - Incremental net monetary benefit	- In-hospital all-cause mortality - 1-year all-cause mortality - Days alive and out of hospital at 30-days and 1-year - Discharge destination - Complications - Readmissions or reinterventions within 30-days and 1-year - Incremental QALY - Incremental net monetary benefit
Causal contrasts	Intention to treat (ITT) - Analysing as randomised Per-protocol (PP) effect - Analysing as type of anaesthesia actually received For both, we are interested in the causal contrasts in the overall study population (the average treatment effect) and in clinically important subgroups (conditional average treatment effects) Subgroups include: - Sex - Age group - Frailty status - Systolic blood pressure at baseline - Prevalence of other key comorbidities (chronic lung disease, chronic kidney disease, ischaemic heart disease) - Smoking status - Case severity according to the Hardman index	Per protocol effect - Analysing according to the type of anaesthesia actually received, assuming any person who switched from local to general anaesthesia during surgery was recorded as being in the general anaesthesia group Similar to the target trial, we are interested in the causal contrasts in the overall study population (the average treatment effect) and in clinically important subgroups (conditional average treatment effects). Subgroups will be defined using the NVR-HES-ONS linked data, and will include: - Sex - Age group - Frailty status - Systolic blood pressure at baseline - Prevalence of other key comorbidities (chronic lung disease, chronic kidney disease, ischaemic heart disease) - Smoking status - Case severity according to the Hardman index, which we will approximate with 4 out of the 5 variables needed to calculate this index (without conscious status which is not recorded in the NVR)
Identifying assumptions	Exchangeability	Exchangeability

	- We assume that the randomisation is sufficient so that the treatment and comparator groups are on average comparable on both observed and unobserved characteristics Positivity - Each participant has a non-zero probability of being assigned to the treatment or comparator No differential loss-to-follow-up - People remain in the trial and losses to follow-up are not differential according to treatment arm	- We assume that the instrument in the instrumental variable analysis is not associated with unmeasured confounders, meaning our analysis yields treatment effect estimates unbiased by confounding Other instrumental variable assumptions: - Relevance (the instrument strongly predicts the surgery actually received) - Exclusion restriction (the instrument acts on the outcome only via the surgery received) - Monotonicity (the instrument has a consistent impact on the surgery received) Positivity - Each participant has a non-zero probability of receiving the treatment or comparator (achieved through strict eligibility criteria developed by a clinical panel and confirmed via baseline characteristics) - The instrument varies across strata of the covariates No differential loss-to-follow-up - NVR-HES-ONS data capture is sufficient in that there is no differential missingness in outcome data over follow-up
Data analysis plan	**Descriptive** Baseline characteristics are compared across treatment groups to ensure randomisation balanced observed patient characteristics. The crude outcomes will be reported as proportions or rates stratified by treatment arm, and by clinically important subgroups of interest. **Clinical effectiveness** Generalised linear models (binary or count outcomes) or survival analysis using Cox proportional hazards regression (time-to-event outcomes) will be used, adjusting for measured	**Descriptive** Baseline characteristics are compared across treatment groups to understand potential confounding in the eligible study population. Variation in the 'tendency to expedite surgery' (proportion of people with 'urgent' surgery in the 6-months prior to the index date for each individual patient) will be compared across vascular centres in England to explore exogenous variation in instrument.

	covariates to adjust for chance imbalances between treatment groups and to improve precision. We will report the relative risk and absolute differences in the probability of death at 30-days follow-up (primary outcome) and other secondary outcomes. The absolute mean difference will be reported for DAOH, and absolute risk difference for binary endpoint measures. We will report the clinical effectiveness overall, and according to pre-specified subgroups. For count data (e.g., DAOH), the outcome analysis will use generalised linear models (e.g., Poisson models). Probit or logit models will be used for binary outcomes (30-day, 90-day, and 1-year mortality and emergency admission). A discrete-time hazard formulation will be used for survival. **Cost-effectiveness** Generalised linear models with appropriate distributions and link functions based on the nature of the data, adjusting for chance imbalances in covariates between treatment groups will be used to calculate key cost-effectiveness analysis outcomes (life years, QALYs, total costs, and incremental net monetary benefit).	We will partially evaluate the instrumental variable assumptions by testing how strongly the instrument predicts treatment and the balance of measured confounders across levels of the instrument. **Clinical effectiveness** We will conduct the instrumental variable analysis using a two-stage model instrumental variable model (two stage residual inclusion or the local IV model). We will report the relative risk and absolute differences in the probability of death at 30-days follow-up (primary outcome) and other secondary outcomes. The absolute mean difference will be reported for DAOH, and absolute risk difference for binary endpoint measures. We will report the clinical effectiveness overall, and according to pre-specified subgroups. For count data (e.g., DAOH), the outcome analysis will use generalised linear models (e.g., Poisson models). Probit or logit models will be used for binary outcomes (30-day, 90-day, and 1-year mortality and emergency admission). A discrete-time hazard formulation will be used for survival. Multivariable regression will be used to improve precision of estimates **Cost-effectiveness** We will calculate total costs by treatment arm using unit costs from a literature review, the NHS payment by results database, and the PSSRU unit cost database. We will also report quality adjusted life years (QALYs) by treatment arm by combining survival time with quality of life estimates using the area under the curve approach. We will extend the previous instrumental variable analysis to report incremental QALYs and costs of local vs general

		anaesthesia. The cost-effectiveness will be reported according to incremental net monetary benefit.

AAA, abdominal aortic aneurysm; EVAR, endovascular aneurysm repair; HES, Hospital Episode Statistics; NVR, National Vascular Registry;

Table 2. Key instrumental variable assumptions relevant to the ELATE study*

Assumption	Details	Application in ELATE
Relevance	The instrument strongly predicts the type of anaesthetic actually received. *Testable by calculating the F-statistic summarising the strength of association between the instrument and the observed anaesthetic type.*	People who are managed at hospitals with a higher 'tendency to use local vs general anaesthetic' have a higher propensity to receive local versus general anaesthetic.
Exchangeability	The instrument is not associated with unmeasured confounders. *Partially testable. We can only observe associations between the instrument and measured confounders. The absence of any associations will increase our confidence (but will not confirm) that the instrument is not associated with unmeasured confounders.*	The 'tendency to use local vs general anaesthetic' is independent of unmeasured confounders which bias the observed associations between local versus general anaesthetic and the study outcomes.
Exclusion restriction	The instrument affects the outcome only through the type of anaesthetic (not via any other pathway). *Not testable. We will use directed acyclic graphs informed by previous research and clinical judgement to justify this assumption.*	There is no independent path leading from the instrument to the outcomes of interest.
Monotonicity	The instrument induces changes in the type of anaesthetic used in one direction only for all individuals. *Not testable. We cannot observe counterfactual anaesthetic decisions for the same individual across different vascular centres with different levels of the instrument.*	We assume that there are no defiers in the study population (i.e., that there are no patients that would get local anaesthetic when not encouraged by the instrument, but would get general anaesthetic when encouraged to get local anaesthetic).

*Table adapted from the ESORT-V protocol (unpublished)
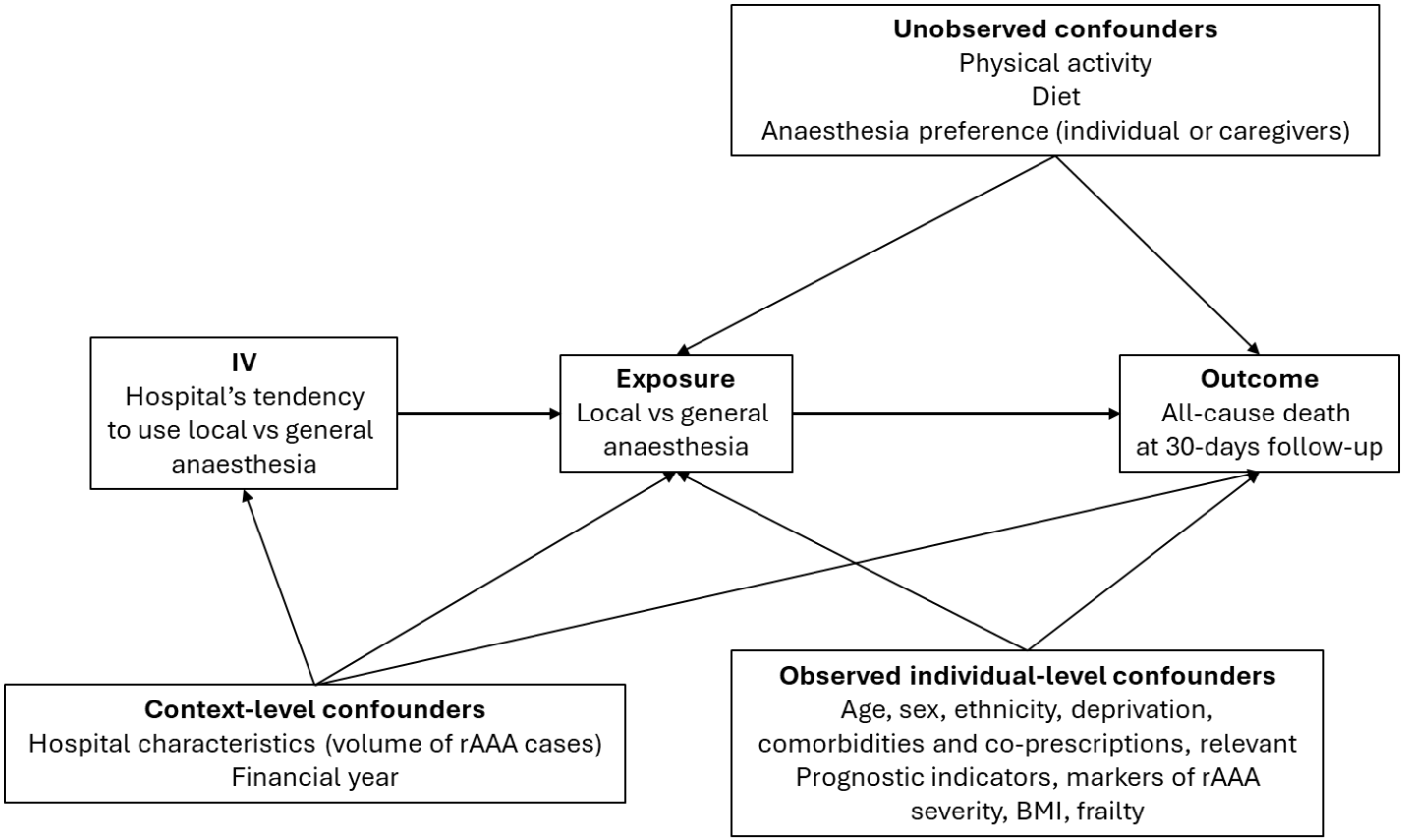
Figure 1. Conceptual diagram illustrating the causal framework for the ELATE study